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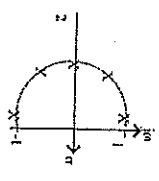
III SEMESTER B.Tech (ECE, EIOT)
END-SEMESTER EXAMINATION 2023

Course Code- ECECC05 / EIECC05
Time: 3:00 Hrs

Course Title- Signals & Systems
Max Marks: 50

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q.N.	Questions	Marks	CO
Q1	Attempt any 2 out of 3 parts		
a	Check whether the following system is linear, time-invariant, causal, stable and memory less. $y(n) = (-1)^n x(n)$.	05	CO 1
b	Consider the signal $x(t)$. $x(t) = \begin{cases} t & 0 \leq t \leq 1 \\ -t + 2 & 1 < t \leq 2 \\ 0 & \text{otherwise} \end{cases}$ Represent the given signal in terms of (i) unit-ramp signal, (ii) unit-step signal, and plot its derivative.	05	
c	Determine whether the following signals are power signals or energy signals or neither. (i) $x[n] = j^{\frac{n}{4}}$ (ii) $x[n] = (\sqrt{j})^n + (\sqrt{j})^{-n}$	05	
Q2	Attempt any 2 out of 3 parts		
a	Let $x[n]$ be the input to an LTI system with and impulse response $h[n]$. Given $x[n] = \delta[n] + 2\delta[n-1] - \delta[n-3]$ $h[n] = 2\delta[n+1] + 2\delta[n-1]$. (i) Compute $Y[n] = x[n] * h[n+2]$. (ii) Express $Y[n]$ in terms of the step response of the LTI system.	05	CO 2
b	Compute $x(t) * y(t)$, where $x(t) = \begin{cases} 1, & \text{for } t \leq 1 \\ 0, & \text{elsewhere} \end{cases}$ and $y(t) = \begin{cases} 1, & \text{for } 0 < t \leq 2 \\ -1, & \text{for } -2 < t \leq 0 \\ 0, & \text{elsewhere} \end{cases}$. Also plot $x(-t) * y(-t)$.	05	
c	Let $x[n] = \begin{cases} 1, & 0 \leq n \leq 9 \\ 0, & \text{e.w.} \end{cases}$ and $h[n] = \begin{cases} 1, & 0 \leq n \leq N \\ 0, & \text{e.w.} \end{cases}$ Where, $N \leq 9$ is an integer. Determine the value of N , given that $y[n] = x[n] * h[n]$ and $y[4] = 5$, $y[14] = 0$.	05	

Q3 Attempt any 2 out of 3 parts			
a	Consider a sequence $x(n)$ and its DTFT $X(e^{j\omega})$. The following facts are known. (i) $x(n)$ is real and causal (ii) $X(e^{j\omega}) = \frac{5}{4} - \cos(\omega)$. Determine a sequence $x(n)$ consistent with the given information.	05	CO 3
b	Suppose you have the Fourier series of two periodic signals $x(t)$ and $y(t)$ of periods T_1 and T_2 , respectively. Let X_n and Y_n be the Fourier series coefficients corresponding to $x(t)$ and $y(t)$. If $T_1 = 2T_2$, determine the Fourier series coefficients of $z(t) = x(t) + y(t)$ in terms of X_n and Y_n . Consider the following signal. $x(t) = [1 + \cos(\pi t)]u(t+1) - u(t-1)$. Find the Fourier transform of $x(t)$ and plot its magnitude.	05	
c	Find the Fourier transform of $x(t)$ and plot its magnitude.	05	
Q4 Attempt any 2 out of 3 parts			
a	i. Compute the Laplace Transform of $x(t) = te^{-3t}u(t+1)$ and specify the ROC, using Laplace transform properties. ii. Let $X(s)$ denote the Laplace transform of $x(t)$. It is given that $x(t)$ is absolutely integrable and $X(s)$ is a rational function, with a pole at $s=-1$, in addition to other unknown poles. Determine if $x(t)$ can be Left Sided, Right Sided, Two Sided, Finite Duration.	2.5+ 2.5	CO4
b	i. Suppose that the LTI system described by $H(s) = \frac{2}{(s+3)(s-1)}$ is known to be stable. Is this system causal? Compute its impulse response $h(t)$. ii. Sketch the magnitude and phase response of the system described by $H(s) = \frac{s-1}{s+2}$. Find the ROC and $h(t)$.	2.5+ 2.5	
c	i. $X(s) = \frac{2s+3}{s^2+3s+2}$ and $Y(s) = \frac{3s+1}{s^2+4s+3}$ are the Laplace transforms of the two sided signals $x(t)$ and $y(t)$. Find all possible $z(t)$, if $Z(s) = X(s) + Y(s)$. ii. The pole zero plot of $X(s)$ is as given below. Is the signal $x(t)$ real, even, oscillatory?	2.5+ 2.5	
			

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Q5 Attempt any 2 out of 3 parts			
a	If the input to an LTI system is $x[n] = u[n]$, the output is $y[n] = (\frac{1}{2})^{n-1}u[n+1]$. i. Find $H(z)$ and the pole zero plot. ii. Compute $h[n]$. iii. Is the system stable? iv. Is the system causal?	05	CO5
b	Use Z-transform to evaluate the following (i) $\sum_{n=0}^{\infty} (0.3)^n$ (ii) $\sum_{n=5}^{\infty} (0.3)^n$ (iii) $\sum_{n=0}^{\infty} (0.5)^n \cos(0.1n)$	05	
c	A signal $y[n]$ is related to another signal $x[n]$ by $y[n] = \sum_{m=-\infty}^n x[m]$. If $y[n] \leftrightarrow \frac{1}{(2-z)^2}$, $ z > 1$. What are the numerical values of $x[-1]$, $x[0]$, $x[1]$ and $x[-2]$	05	

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